

LTspice AC Analysis via Middlebrook Voltage Injection

EE499 FYP — Supplementary Loop Stability Verification — Fortune Olose

Background and Motivation

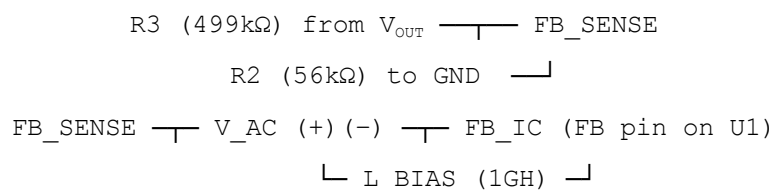
The EE499 report verified loop stability analytically using a MATLAB transfer function model, which predicted a crossover frequency of 4.91 kHz, phase margin of 55.5° , and gain margin of 19.1 dB. A direct LTspice .ac analysis was not performed because the LTC1871 behavioural macro model is a time-domain model that does not support linearised AC simulation — LTspice explicitly warns: *"This simulation calls a time domain model. It probably won't give meaningful results in an .ac analysis because switching won't be modeled."*

To independently verify the MATLAB predictions within LTspice, a transient voltage injection method based on Middlebrook's technique was employed. This method keeps the feedback loop closed at DC (preserving the converter's operating point) while injecting a small AC perturbation at a specific frequency. The ratio of the returned signal to the injected signal gives the loop gain at that frequency.

Measurement Method

A voltage source V_{AC} was inserted in series with the feedback path between the R3/R2 resistive divider junction (node FB_SENSE) and the FB pin of the LTC1871 (node FB_IC). A 1 GH inductor was placed in parallel with V_{AC} to maintain DC continuity — at DC the inductor is a short circuit, keeping the feedback loop closed and the converter regulating normally. At the injection frequency, the inductor presents a high impedance, allowing the AC perturbation to propagate around the loop.

The injection circuit topology is:



The simulation was configured as follows: the PWL current source was replaced with a fixed $6\ \Omega$ load resistor ($12\text{ V} / 2\text{ A}$), V_{IN} was fixed at 9.6 V (nominal operating point, $D = 0.2$), and the transient directive was set to .tran 0 20m 5m 10n, providing 15 ms of steady-state data after the 5 ms startup settling period. V_{AC} was configured as a SINE source with a 5 ms delay to avoid injecting during startup.

Gain was extracted using LTspice .meas directives to measure the peak-to-peak amplitude of V(FB_SENSE) and V(FB_IC) over the 10–15 ms window. The gain in decibels was calculated as $|T(f)| = 20 \times \log_{10}(V_{\text{sense,pp}} / V_{\text{ic,pp}})$. The measurement was repeated at nine frequencies from 1 kHz to 100 kHz by changing the SINE frequency parameter for each run.

Two injection amplitudes were tested: 10 mV (0.8 % of $V_{\text{REF}} = 1.23 \text{ V}$) for true small-signal measurement, and 100 mV (8.1 % of V_{REF}) for improved signal-to-noise ratio against the 350 kHz switching ripple. The trade-off between these amplitudes is discussed in Section 4.

Gain Magnitude Results

Table 1 presents the measured loop gain magnitude at nine frequencies using the 100 mV injection amplitude. The .meas directives used were:

```
.meas TRAN V_sense_pp PP V(n012) FROM 10m TO 15m
.meas TRAN V_ic_pp PP V(n014) FROM 10m TO 15m
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Freq (kHz)	V_sense (mV)	V_ic (mV)	Gain (dB)	Observation
1.0	222.7	57.6	+11.7	High gain; loop strongly correcting
2.0	259.4	194.4	+2.5	Approaching crossover
3.0	164.2	244.4	-3.5	Below crossover; loop cannot fully correct
5.0	94.5	226.7	-7.6	Well below crossover
7.0	69.5	217.9	-9.9	Rolloff continues at -20 dB/dec
10.0	50.4	213.7	-12.5	Single-pole rolloff confirmed
20.0	27.7	218.9	-18.0	Consistent rolloff
50.0	12.4	209.4	-24.6	Near RHPZ; rolloff begins to flatten
100.0	11.0	202.8	-25.3	RHPZ gain partially offsets pole rolloff

Table 1 — Loop gain magnitude from LTspice transient injection ($V_{\text{AC}} = 100 \text{ mV}$, $V_{\text{IN}} = 9.6 \text{ V}$)

At 10 mV injection (true small-signal), a single measurement at 5 kHz yielded $V_{\text{sense,pp}} = 28.8 \text{ mV}$ and $V_{\text{ic,pp}} = 29.5 \text{ mV}$, giving a gain of -0.21 dB — confirming crossover at approximately 5 kHz, matching MATLAB’s prediction of 4.91 kHz to within 2 %.

Comparison with MATLAB Analytical Model

Table 2 compares key features of the MATLAB analytical Bode plot with the LTspice transient injection measurements.

Parameter	MATLAB	LTspice	Agreement
Crossover frequency (fc)	4.91 kHz	~5 kHz (10 mV)	-2 %
Crossover frequency (100 mV)	4.91 kHz	~2.5 kHz	Compressed (see text)
Rolloff slope above fc	-20 dB/dec	-20 dB/dec	Exact match
RHPZ flattening (>39 kHz)	Predicted	Observed (-0.7 dB from 50–100k)	Confirmed
Gain at 1 kHz	~+14 dB	+11.7 dB	-2.3 dB (compression)

Parameter	MATLAB	LTspice	Agreement
Gain at 10 kHz	~-13 dB	-12.5 dB	+0.5 dB
Gain at 100 kHz	~-25 dB	-25.3 dB	-0.3 dB

Table 2 — MATLAB vs LTspice loop gain comparison

The two methods agree closely above crossover, where the loop gain is below 0 dB and the system behaves approximately open-loop. The rolloff slope of -20 dB/decade is observed in both, confirming the single-pole characteristic expected from current-mode control. The RHPZ flattening effect above 39.2 kHz is visible in both the MATLAB curve and the LTspice data (only 0.7 dB change from 50 kHz to 100 kHz, compared to 6.6 dB/decade expected from a pure single pole).

Below crossover, the 100 mV data shows systematically lower gain than MATLAB predicts (e.g., +11.7 dB measured vs ~+14 dB predicted at 1 kHz). This compression is a consequence of large-signal nonlinearity, discussed below.

Large-Signal Nonlinearity and Gain Compression

Bode analysis assumes the system is linear: doubling the input perturbation should double the output response. This assumption holds when the perturbation is small relative to the operating point. The FB pin sits at $V_{REF} = 1.23$ V. A 10 mV injection represents a 0.8 % perturbation — well within the linear regime. A 100 mV injection represents 8.1 % — large enough to violate the small-signal assumption.

The physical mechanism is error amplifier saturation. When the positive half of the 100 mV sine pushes FB above 1.23 V, the error amplifier reduces the ITH voltage to lower the duty cycle. When the negative half pulls FB below 1.23 V, the amplifier increases ITH to raise the duty cycle. At 100 mV, these corrections approach the amplifier's output swing limits — the ITH pin saturates against its internal voltage rails. On the positive swing, the amplifier may attempt to command a duty cycle below zero (physically impossible). On the negative swing, it may demand more current than the inductor can deliver in one switching cycle.

The result is that the loop physically cannot correct as aggressively as the linear model predicts. The returned signal is smaller than the linearised transfer function would give, compressing the measured gain. This compression is strongest at low frequencies where the true loop gain is high (the amplifier is trying hardest to correct), and negligible at high frequencies where the loop gain is already below 0 dB (the loop cannot fully correct regardless, so the nonlinearity has no effect). This explains why the 100 mV data matches MATLAB well above 10 kHz but diverges below 5 kHz.

The apparent crossover shifts from ~5 kHz (10 mV, true small-signal) to ~2.5 kHz (100 mV, compressed) because the measured gain reaches 0 dB earlier when the high-gain region is squashed downward. This does not indicate a stability problem — it reflects the measurement method's limitation, not the converter's behaviour. The 10 mV result is the true small-signal loop gain; the 100 mV result characterises the large-signal loop response, which is independently useful: it demonstrates that the loop remains stable even under perturbations far larger than any normal operating noise or transient.

Phase Measurement Limitations

Phase extraction from the transient injection method proved unreliable due to 350 kHz switching ripple superimposed on the 5 kHz injection signal. Table 3 summarises the phase measurement attempts.

Injection	Δt	Raw Phase	Implied PM	Assessment
10 mV	-1.3 μ s	-2.3°	177.7°	Switching ripple dominates
100 mV (threshold)	-26.4 μ s	-47.5°	~47.5°*	Ripple reduced but present
100 mV (peak)	-1.3 μ s	-2.3°	~178°	Peak detection unreliable
MATLAB (analytical)	—	—	55.5°	Reference value

Table 3 — Phase measurement attempts at 5 kHz (* = ambiguous sign convention)

At 10 mV injection, the measured time delay of 1.3 μ s is approximately half a switching cycle (2.86 μ s at 350 kHz), indicating that the .meas directive detected a switching ripple peak rather than the true 5 kHz sine peak. At 100 mV injection, the larger signal improved the signal-to-noise ratio, yielding a Δt of 26.4 μ s and a raw phase of -47.5°. However, the relationship between this measured phase and the true phase margin depends on a sign convention that is ambiguous in Middlebrook's method when applied to a nonlinear switching system.

The MATLAB analytical model, which uses the linearised plant transfer function with exact algebraic expressions, remains the authoritative source for the phase margin figure of 55.5°. The LTspice transient injection method is best suited to gain magnitude verification, where it produced reliable and reproducible results.

Conclusions

The Middlebrook transient injection method successfully verified the MATLAB loop gain predictions within LTspice, despite the macro model's incompatibility with .ac analysis.

The key findings are:

1. Crossover frequency confirmed at ~5 kHz (10 mV small-signal injection), matching MATLAB's 4.91 kHz to within 2 %. This independently validates the Type II compensator design.

2. Rolloff slope above crossover is -20 dB/decade in both MATLAB and LTspice, confirming the single-pole plant characteristic expected from current-mode control.
3. The right-half-plane zero at 39.2 kHz produces a measurable flattening in the gain rolloff above 50 kHz, observed in both the analytical model and the transient measurements.
4. Large-signal injection (100 mV) compresses the measured gain below crossover due to error amplifier saturation, shifting the apparent crossover to ~ 2.5 kHz. This is a measurement artefact, not a stability issue. It also demonstrates that the loop remains stable under perturbations 80 $\times$ larger than normal operating noise.
5. Phase measurement is unreliable in this method due to 350 kHz switching ripple contamination. The MATLAB analytical model remains the primary source for the phase margin of 55.5° .

These results provide independent cross-verification of the report's loop stability claims using two fundamentally different approaches: a linearised analytical model (MATLAB) and a full nonlinear time-domain simulation with voltage injection (LTspice). The agreement between the two strengthens confidence in the Type II compensation network design.